Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We consider the problem of covering/connecting a multi-interface network with a minimum cost assignment of interfaces to vertices. In the Coverage problem we demand that every edge is covered by at least one common interface assigned to its endpoints. In the Connectivity problem we are additionally given a set of disjoint groups of terminals $U_1, \ldots, U_p \subseteq V$ and we demand that the subgraph of activated edges spans each group U_i. The cost of activating an interface at a vertex is given by a non-negative function $c\colon V \times [k] \to \mathbb{N}$, where k is the number of interfaces. We wish to minimize the maximum cost of an activated interface.

For the Coverage problem we give the first $O(\log m)$-approximation algorithm based on a randomized rounding of a natural, set-cover-style linear programming relaxation. We also show how to use this LP to obtain a k-approximation algorithm, where k is the number of interfaces. This improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation, where b is a certain parameter of the input instance which can be as large as $\Omega(n)$ and $c_{\max}$ is the maximum cost of activating an interface and $O(k)$-approximation for the uniform cost case [Algorithmica 2012]. For the Connectivity problem we devise an $O(\log^2 m)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity of all terminal groups. The rounding procedure is similar to the one for the Coverage problem but requires a more careful analysis to ensure that the connectivity constraints are satisfied with high probability.

1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by choosing between large variety of interfaces available at each device. Moreover, the devices present in the network may vary according to the interfaces they provide for the usage. For example some device may be able to connect by Bluetooth and Wi-Fi, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as a graph problem. Every device in the network is modeled by a vertex with a set of labels representing its available interfaces. Two devices can communicate if they share a common available interface and are sufficiently close to each other, and the edges of the graph represent the possible connections. A connection is established between two devices when they share a common active interface. We want to find an assignment of active interfaces so that we ensure either Coverage or Connectivity. In the Coverage problem, we wish to establish every possible connection in the network (see Figure 1 for an example) while in the Connectivity problem, we are additionally given a collection of disjoint vertex subsets

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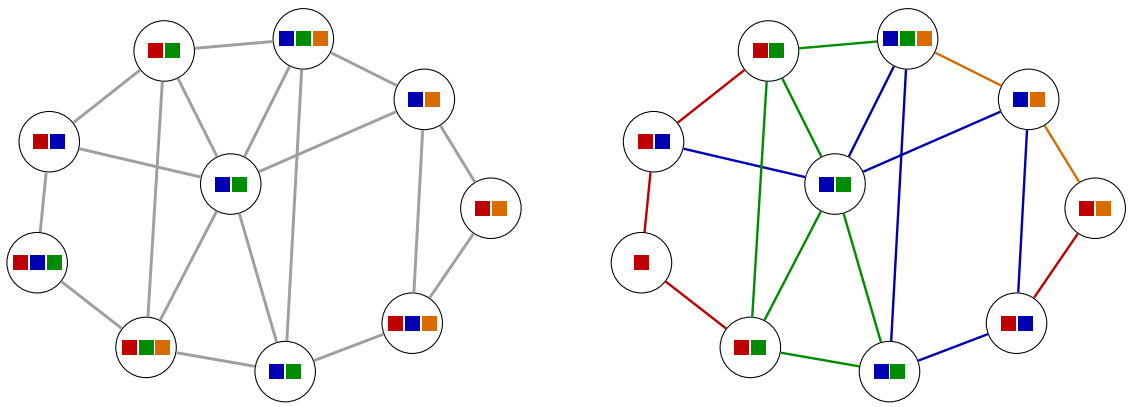
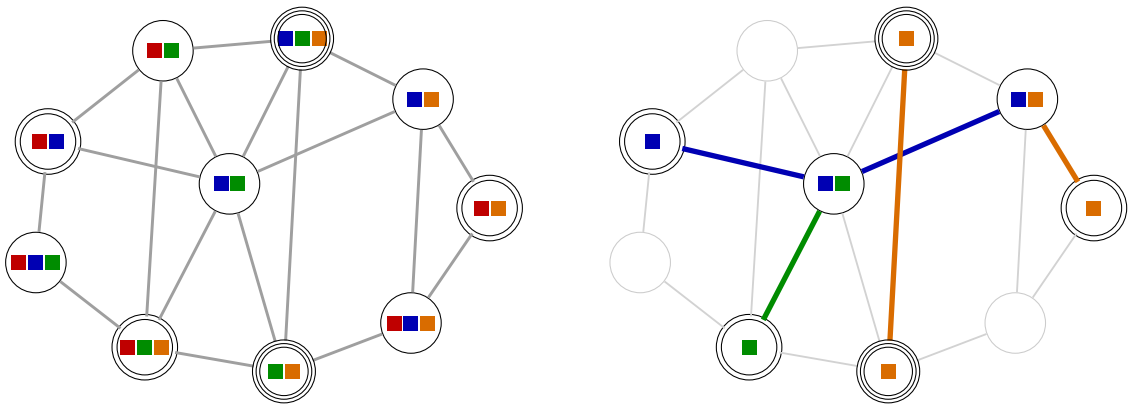


Available labels

Example covering assignment

Figure 1: An example instance of a multi-interface network with available interfaces at each vertex and a feasible covering assignment.



Available labels (double/triple contour = terminal group 1/2)

Example connecting assignment

Figure 2: An example instance of a multi-interface network with available interfaces at each vertex and a feasible connecting assignment.

called terminals, and we require each terminal group to be connected in the subgraph induced by active interfaces (see Figure 2 for an example). In both problems, we assume that activating an interface induces a cost that depends on the type of interface and on the vertex that activates it. The goal is to minimize the min-max cost, i. e., the maximum cost of activated interfaces across all vertices.

1.1 Related Work

In the following, we denote k the number of different interfaces across the network, n the number of vertices, Δ the maximum degree of the graph, $c_{\max}$ the maximum cost and $c_{\min}$ the minimum cost of an interface. In the *homogeneous* setting, the costs depend on the interface only while in the *heterogeneous* case they depend on the interface and on the vertex.

Coverage Problem. The algorithmic study of problems in multi-interface networks started with the Coverage problem, introduced in Caporuscio et al. [6]. The Coverage problem has been studied only in the homogeneous setting, both for min-sum [11] and min-max [7] objective functions.

The min-sum variant is APX-hard for any constant $k \geq 3$ and unit costs, and it is hard to approximate within a $O(\ln k)$ factor even with unit costs, but it has a $\min\{k-1, \sqrt{n}(1+\ln n)\}$-approximation [11].

The min-max variant is NP-Hard for any constant $\Delta \geq 5$, $k \geq 16$ and for unit costs, and it is not approximable within a $\eta \ln \Delta$ factor for some constant η even on trees and with unit costs. On the positive side, it admits a $(\ln \Delta + 1 + b \cdot \min\{c_{\max}, \ln \Delta + 1\})$-approximation, where b is a parameter that depends on the structural properties of the input graph and is bounded by many other graph structutal parameters such as treewidth, arboricity or maximum degree. It also has a $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$-approximation and a $(\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}})$-approximation [7].

Connectivity Problem. The Connectivity problem was introduced by Kosowski et al. [13]. It was first studied in the homogeneous setting and in the special case where all vertices are terminals, both for the min-sum [13] and the min-max objective functions [7].

The min-sum variant with terminals only and homogeneous costs is APX-hard for $k \geq 2$, $\Delta \geq 4$ and unit costs already, but has a 2-approximation [13]. This result was improved by Athanassopoulos et al. [4] to a randomized $(\frac{3}{2} + \epsilon)$-approximation for any constant ϵ. Moreover, this approximation algorithm applies to the more general problem where for each interface, we are additionally given the set of edges that it can cover.

The min-max variant where costs are homogeneous and all vertices are terminals is NP-Hard for any constant $\Delta \geq 3$, $k \geq 10$ and for unit costs. Similarly to the Coverage problem, it is not approximable within a $\eta \ln n$ factor even for unit costs, but has simple $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$- and $\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}}$-approximations [7].

Athanassopoulos et al. [4] expand the study of the min-sum Connectivity problem to *group connectivity*, a more general problem where the solution graph should connect together groups of vertices that might not span the whole set of vertices. They provide a randomized 4-approximation for it. They also initiate the study of this problem in the heterogeneous setting, where costs can also depend on the vertex. They show that the heterogeneous Connectivity problem is hard to approximate within $o(\ln n)$, but the group connectivity heterogeneous case admits a $O(\ln n)$-approximation.

Other Multi-interface Problems. There has been some work on the study of the Coverage problem from the perspective of parameterized complexity, possibly with an addtional constraint on the maximum number of active interfaces in a vertex [2, 1, 5], yielding efficient algorithms in some graph classes. Some papers also study other problems in multi-interface networks, like cheapest path [13], matchings [12], or flow problems [8].

1.2 Our contribution and techniques

We study the min-max Coverage and Connectivity problems in the heterogeneous setting, where activation costs may depend on both the interface and the vertex. We give a set-cover-style LP relaxation for the min-max Coverage problem and show that treating the resulting fractional solution as a probability distribution and applying a randomized rounding procedure yields an $O(\log m)$-approximation, where m is the number of edges in the graph. This is the first logarithmic approximation for this problem, and it matches the lower bound of $\Omega(\log n)$ even for unit costs and on stars [7]. This improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation of [7], which was limited to the homogeneous case. A simpler deterministic rounding of the same LP gives a k-approximation, where k is the number of interfaces.

For the Connectivity problem we extend the LP by adding flow-based cut constraints encoding group connectivity. Our rounding procedure is similar to the one for the Coverage problem, however, this time our rounding is iterative and the algorithm proceeds until all terminal groups are connected. Each iteration is randomized and we show that in expectation in each such round the algorithm covers a siginificant fraction of edges of some graph which connects

all group terminals with high probability. This yields an $O(\log^2 m)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $[k]$ be the set of different types of interfaces across the network and let $\lambda\colon V \to [k]$ be a function that assigns to each vertex $v \in V$ its set of available interfaces such that for each edge $uv \in E$, $\lambda(u) \cap \lambda(v) \neq \emptyset$. An assignment of active interfaces is a function $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$. We say that an edge $uv \in E$ is *covered* if $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$. We denote by $\mathrm{cov}(\lambda_A) = \{uv \in E : \lambda_A(u) \cap \lambda_A(v) \neq \emptyset\}$ the set of edges covered by λ_A, and by G_A the subgraph of G with edge set $\mathrm{cov}(\lambda_A)$.

In the Coverage problem, we wish to find an assignment of active interfaces λ_A such that $G_A = G$. We call such an assignment a *covering assignment*.

In the Connectivity problem, we are additionally given $p \geq 1$ pairwise disjoint groups of *terminals* $U_1, U_2, \ldots, U_p \subseteq V$ and we wish to find an assignment of active interfaces λ_A such that for every $r \in [p]$, all vertices in U_r lie in the same connected component of G_A. Vertices in $V \setminus \bigcup_{r=1}^{p} U_r$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $p = 1$ with $U_1 = V$ corresponds to requiring that G_A is a connected spanning subgraph of G.

We are also given a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, which denotes the cost of activating the interface $i \in [k]$ in the vertex $v \in V$. For an assignment λ_A, we define the *cost at vertex* v as:

$$\mathrm{cost}(v, \lambda_A) = \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

Based on this, we consider two global cost criteria. The *max cost* of an assignment λ_A is:

$$\mathrm{cost}_M(\lambda_A) = \max_{v \in V} \mathrm{cost}(v, \lambda_A)$$

and the *sum cost* of an assignment λ_A is:

$$\mathrm{cost}_S(\lambda_A) = \sum_{v \in V} \mathrm{cost}(v, \lambda_A)\,.$$

The goal of the min-max (resp. min-sum) assignment problem is to find an assignment λ_A that minimizes cost_M (resp. cost_S). The considered problems are formalized below:

Coverage Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$.
Task: Find a min-max (resp. min-sum) covering assignment λ_A minimizing $\mathrm{cost}_M(\lambda_A)$ (resp. $\mathrm{cost}_S(\lambda_A)$).

Connectivity Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, pairwise disjoint groups of terminals $U_1, \ldots, U_p \subseteq V$.
Task: Find a min-max (resp. min-sum) connecting assignment λ_A minimizing $\mathrm{cost}_M(\lambda_A)$ (resp. $\mathrm{cost}_S(\lambda_A)$).

Throughout this paper, unless stated otherwise, we focus on the *min-max* cost criterion. Therefore, we will simply write $\mathrm{cost}(\lambda_A)$ instead of $\mathrm{cost}_M(\lambda_A)$ and we will refer to the min-max assignment problem as the Coverage and Connectivity problems, respectively. We will explicitly state when we consider the min-sum cost criterion.

2.1 Probabilistic tools

Additionally, we will need the following probability facts:

Theorem 2.1 (Markov's Inequality)**.** *Let X be a non-negative random variable and $a > 0$. Then:*

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 2.2 ([15] Wald's Identity)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite expectation $\mathbb{E}[X]$, and let N be a non-negative integer-valued random variable independent of the sequence $(X_i)_{i\geq 1}$ with finite expectation $\mathbb{E}[N]$. Then:*

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N] \cdot \mathbb{E}[X].$$

Theorem 2.3 ((weighted) Chernoff Bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0, 1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0, 1]$. Let $X = \sum_{i=1}^{n} w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0, 1)$ we have that:*

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1-\delta_2)\cdot\mu) \leq \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

The proof of the above fact follows the proof of the standard Chernoff bound and we include it in the appendix for completeness.

3 $O(\log n)$-approximation algorithms for the Coverage problem

Below we show an $O(\log m)$ approximation algorithms for all four variants and $O(k)$-approximation for the Coverage problem. The idea of our ILP formulation and the algorithm is inspired by the rounding procedure known for set cover [14] and related problems [3]

3.1 Rounding the costs

Due to technical reasons, we show how to produce a polynomial number of instances, with cost function bounded by a polynomial in n. By testing consecutive powers of 2, we find the smallest number b, such that the costliest activated interface in the optimal solution has cost at most 2^b. For each vertex $v \in V$, we discard all of the interfaces from $\lambda(v)$ with cost larger than 2^b and we normalize the costs of remaining interfaces by dividing them by 2^b. Note that the number of guesses is polynomial. We also can assume that $\texttt{OPT} \geq 1/2$, since the cost of any solution is at least the cost of activating one interface, which is at least $1/2$ by our rounding.

3.2 ILP formulation of covering multi-interface networks

We start with the following ILP formulation of the min-sum covering assignment problem. Let $x_{i,v}$ be a binary variable that is 1 if and only if the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. We consider two types of vertices. For a given vertex $v \in V$, if $\sum_{i\in\lambda(v)} c(i, v) \leq 1$, then we can safely assume that all interfaces in $\lambda(v)$ are activated at v, since the cost of activating all of them is at most 1 and thus at most $2 \cdot \texttt{OPT}$. We call such vertices *cheap*. For the remaining vertices, we call them *expensive*. Note that for every expensive vertex v, we can assume that

$\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)\geq 1$, since this will always result in a solution of cost at most $2\cdot \texttt{OPT}$. Thus, we can write the following ILP formulation of the min-max covering assignment problem:

$$\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)\leq M \quad \forall v\in V \\
& \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u},x_{i,v}\}\geq 1 \quad \forall uv\in E \\
& \begin{cases} x_{i,v}=1 \quad \forall i\in\lambda(v) & \text{if } \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v}<1 \\ \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v}\geq 1 & \text{otherwise} \end{cases} \quad \forall v\in V \\
& x_{i,v}\in\{0,1\} \quad \forall v\in V, i\in\lambda(v)
\end{aligned}$$

Note that the above formulation is not linear due to the presence of the min function. However, by adding an additional variable $y_{i,uv}$ we can linearize the constraints of type $\sum_{i\in\lambda(u)\cap\lambda(v)}\min\{x_{i,u},x_{i,v}\}\geq 1$ by replacing them with $\sum_{i\in\lambda(u)\cap\lambda(v)} y_{i,uv}\geq 1$ together with the constraints $y_{i,uv}\leq x_{i,u}$ and $y_{i,uv}\leq x_{i,v}$ for every $i\in\lambda(u)\cap\lambda(v)$ and $uv\in E$. Thus, we can solve the LP relaxation of this formulations in polynomial time.

3.3 The algorithm

Algorithm 1: A relaxation-based $O(\log m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}$.
2. Pick independent uniformly random thresholds $t_i\in(0,1)$, for each $i\in[k]$.
3. $\lambda_A(v)\leftarrow\{i\in\lambda(v)\colon 3\ln m\cdot x_{i,v}\geq t_i\}$, for every $v\in V$.
4. **return** $\{\lambda_A(v)\}_{v\in V}$.

Theorem 3.1. *The above algorithm is an $O(\log m)$-approximation algorithm for the min-max covering assignment problems which succeeds with high probability.*

Proof. We have the following lemmas:

Lemma 3.2. *The algorithm returns a solution of cost at most $9\ln m\cdot \texttt{OPT}$ with high probability.*

Proof. Fix a vertex $v\in V$. We want to show that with high probability, the cost of activating interfaces at v is at most $9\ln m$ times the cost of activating interfaces at v in the fractional solution. If v is cheap, then the cost of activating all interfaces at v is at most 1, so the claim holds. Assume that v is expensive. By $X_{v,i}$ denote the random variable that is 1 if $3\ln m\cdot x_{i,v}\geq t_i$ and 0 otherwise. Then we can write $\text{cost}(v,\lambda_A)=\sum_{i\in\lambda(v)} X_{v,i}\cdot c(i,v)$. We have: $\mathbb{E}[X_{v,i}]=\Pr[i\in\lambda_A(v)]=3\ln m\cdot x_{i,v}$. Thus, $\mathbb{E}[\text{cost}(v,\lambda_A)]=3\ln m\cdot\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)$. By applying the Chernoff bound with $\delta=2$ we get that

$$\Pr[\text{cost}(v,\lambda_A)\geq 3\cdot\mathbb{E}[\text{cost}(v,\lambda_A)]]\leq\left(\frac{e^2}{3^3}\right)^{\mathbb{E}[\text{cost}(v,\lambda_A)]}\leq\exp\{-\mathbb{E}[\text{cost}(v,\lambda_A)]\}\leq\frac{1}{m^3}$$

where the last inequality is by using the fact that, $\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)\geq 1$. By applying the union bound over all n vertices we get that with probability at least $1-\frac{n}{m^3}\geq 1-\frac{1}{m}$ (assuming $m\geq 2$) for every $v\in V$, $\text{cost}(v,\lambda_A)\leq 9\ln m\cdot\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)$. Thus, with high probability we have $\text{cost}(\lambda_A)\leq 9\ln m\cdot\max_{v\in V}\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v)=9\ln m\cdot\texttt{OPT}$. □

Lemma 3.3. *With high probability, the above algorithm returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-3 \ln m \cdot \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\{-3 \ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1 - \frac{m}{m^3} = 1 - \frac{1}{m^2}$. □

Combining the above lemmas gives the claim. □

3.4 A k-approximation algorithm for the covering problem

Algorithm 2: A relaxation-based k-approximation algorithm.

1 Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.
2 Set $\lambda_A(v) = \{i \in \lambda(v) \colon x_{i,v} \geq \frac{1}{k}\}$, for every $v \in V$.
3 **return** $\{\lambda_A(v)\}_{v \in V}$.

Theorem 3.4. *The above algorithm is a k-approximation algorithm for both the min-sum and the min-max covering assignment problems.*

Proof. Consider an edge $uv \in E$. By averaging, there exists an interface $i \in \lambda(u) \cap \lambda(v)$ such that $\min\{x_{i,u}, x_{i,v}\} \geq \frac{1}{k}$. Thus, $i \in \lambda_A(u) \cap \lambda_A(v)$ so λ_A is a covering assignment. Moreover, since $x_{i,v} \geq \frac{1}{k}$ for every $i \in \lambda_A(v)$, we have that $c(\lambda_A(v)) = \sum_{i \in \lambda_A(v)} c(i, v) \leq k \cdot \sum_{i \in \lambda_A(v)} x_{i,v} \cdot c(i, v)$. Thus, depending on the cost function, we have $\mathrm{cost}(\lambda_A) \leq k \cdot \max_{v \in V} \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v) = k \cdot \mathtt{OPT}$. □

4 $O(\log^2 n)$-approximation algorithm for the Connectivity problem

4.1 ILP formulation

In this section we again assume that $\mathtt{OPT} \geq 1/2$, the cost of activating any interface is at most 1 and we distinguish between cheap and expensive vertices as before. We use a similar LP to model the Connectivity problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e we add an additional flow variable y_e which is 1 if and only if the edge e is covered by the solution. Connectivity within each group U_r is enforced via cut constraints: for every $r \in [p]$ and every cut $(S, V \setminus S)$ that separates at least one pair of terminals in U_r, the total y-weight of edges crossing the cut must

be at least 1. Then we can formulate the min-max CONNECTIVITY problem as follows:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \le M \quad \forall v \in V \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \ge y_{uv} \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \ge 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& y(\delta(S)) \ge 1 \quad \forall r \in [p],\ \forall S \subset V \colon S \cap U_r \ne \emptyset,\ (V \setminus S) \cap U_r \ne \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}
$$

Observe that the above ILP has exponential number of constraints, but we can solve its LP relaxation in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

4.2 The algorithm

Algorithm 3: A relaxation-based $O(\log^2 m)$-approximation algorithm.

Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}, \{y_{uv}\}_{uv \in E}$.
For every edge $e \in E$, let $p_e = \min\{1, 4 \ln m \cdot y_e\}$.
With probability p_e set $Y_e = 1$, otherwise set $Y_e = 0$.
Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
$\mathcal{S} \leftarrow \emptyset$.
$j \leftarrow 1$.
$\lambda_A \leftarrow \{\emptyset\}_{v \in V}$.
while $\mathrm{cov}(\lambda_A, H) \ne E(H)$ **do**
  Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
  $\lambda_A^j(v) \leftarrow \{i \in \lambda(v) \colon x_{i,v} \ge t_i\}$, for every $v \in V$.
  $\lambda_A \leftarrow \lambda_A^j \cup \lambda_A$.
  $j \leftarrow j + 1$.
return λ_A

Theorem 4.1. *There exists an $O\big(\log^2 m\big)$-approximation algorithm for both the min-sum and the min-max CONNECTIVITY problems which succeeds with high probability.*

Proof. We show that the above algorithm returns a solution of cost at most $O\big(\log^2 m\big) \cdot \texttt{OPT}$ with high probability. We have the following lemmas:

Lemma 4.2. *With high probability, for every $r \in [p]$, the subgraph H connects all terminals in U_r.*

Proof. Fix any $r \in [p]$ and any $S \subset V$ with $S \cap U_r \ne \emptyset$ and $(V \setminus S) \cap U_r \ne \emptyset$. Let $\ell \in \mathbb{N}$, such

that $\ell \le y(\delta(S)) < \ell + 1$. We have:

$$\begin{aligned}
\Pr[|\delta_H(S)| = 0] &= \prod_{e \in \delta(S)} \Pr[Y_e = 0] = \prod_{e \in \delta(S)} (1 - p_e) \\
&\le \prod_{e \in \delta(S)} \exp\{-p_e\} \le \exp\left\{ - \sum_{e \in \delta(S)} \min\{1,\, 4 \ln m \cdot y_e\} \right\} \\
&\le \exp\{-4\ell \ln m\} = \frac{1}{m^{4\ell}}
\end{aligned}$$

where the last inequality uses the LP constraint $y(\delta(S)) \ge 1$ for every U_r-separating cut, so $\sum_{e \in \delta(S)} y_e \ge \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [10, 9]:

Lemma 4.3 (Karger's lemma)**.** *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U_r-separating fractional cut has value at least 1, so there are at most $n^{2\ell}$ cuts that are U_r-separating and have fractional value less than ℓ. Applying the union bound over all groups $r \in [p]$ and all U_r-separating cuts:

$$\begin{aligned}
\Pr\big[\exists\, r \in [p],\, \exists\, U_r\text{-sep. cut } S\colon |\delta_H(S)| = 0\big] &\le \sum_{r=1}^{p} \sum_{\ell=1}^{\infty} \sum_{\substack{S \subset V,\, S \cap U_r \neq \emptyset \\ (V \setminus S) \cap U_r \neq \emptyset,\, y(\delta(S)) < \ell+1}} \Pr[|\delta_H(S)| = 0] \\
&\le p \cdot \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{4\ell}} = p \cdot \sum_{\ell=1}^{\infty} \frac{1}{m^{2\ell}} = \frac{p}{m^2 - 1} \\
&= \frac{p}{(m-1)\,(m+1)} \le \frac{1}{m-1},
\end{aligned}$$

where the last inequality uses $p \le n \le m + 1$, since G is connected. Thus, with high probability, no U_r-separating cut is empty in H for any $r \in [p]$. □

Let j be any operation of the while loop. Let H_j be the subgraph of H that is not covered by λ_A at the beginning of the j-th iteration of the while loop. We wish to show, that in expectation, the solution λ_A produced in the j-th iteration of the while loop, call it λ_A^j, covers a significant fraction of the edges in H_j. We have the following lemma:

Lemma 4.4. *We have* $\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] \ge \frac{1-1/e}{3 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]$.

Proof. Consider any edge $uv \in E(G)$. Observe that $\mathbb{E}[Y_{uv}] \le 4 \ln m \cdot y_{uv}$. By the definition of the LP we get that:

$$\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \ge y_{uv} \ge \frac{\mathbb{E}[Y_{uv}]}{4 \ln m}.$$

Therefore, the probability that uv is uncovered by λ_A is at most:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-\min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}
\end{aligned}$$

Thus we get $\Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \geq 1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}$. We now sum over all edges in H_j to get:

$$\begin{aligned}
\mathbb{E}\left[\left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|\right] &= \sum_{uv \in E(H_j)} \Pr[\lambda_A(u) \cap \lambda_A(v) \neq \emptyset] \\
&\geq \sum_{uv \in E(H_j)} \left\{1 - \exp\left\{-\frac{\mathbb{E}[Y_{uv}]}{4 \ln m}\right\}\right\} \\
&\geq (1 - 1/e) \cdot \sum_{uv \in E(H_j)} \frac{\mathbb{E}[Y_{uv}]}{4 \ln m} \geq \frac{1 - 1/e}{4 \ln m} \cdot \mathbb{E}\left[|E(H_j)|\right]
\end{aligned}$$

where the last inequality is by the linearity of expectation. □

Now, we want to show that the number of iterations of the while loop is at most $O\big(\log^2 m\big)$ with high probability. Let I be the random variable denoting the number of iterations of the while loop. We first bound the expected number of iterations.

Lemma 4.5. $\mathbb{E}[I] \leq \frac{9 \ln^2 m}{1 - 1/e}$.

Proof. Let $\alpha = \frac{1 - 1/e}{4 \ln m}$. By definition, $|E(H_{j+1})| = |E(H_j)| - \left|\mathrm{cov}\left(\lambda_A^j, H_j\right)\right|$. By linearity of expectation and the previous lemma, $\mathbb{E}\left[|E(H_{j+1})|\right] \leq (1 - \alpha) \cdot \mathbb{E}\left[|E(H_j)|\right]$. Unfolding the recurrence we get that, $\mathbb{E}\left[|E(H_j)|\right] \leq (1 - \alpha)^j \cdot m \leq m e^{-\alpha j}$. By Markov's inequality, $\Pr\{|E(H_j)| \geq 1\} \leq m e^{-\alpha j}$. Let $j^* = \left\lfloor \frac{9 \ln^2 m}{1 - 1/e} \right\rfloor$. We have:

$$\begin{aligned}
\mathbb{E}[I] = \sum_{j=1}^{\infty} \Pr[I \geq j] &= \sum_{j=1}^{j^*} \Pr[I \geq j] + \sum_{j=j^*+1}^{\infty} \Pr[I \geq j] \\
&\leq j^* + \sum_{j=j^*+1}^{\infty} m e^{-\alpha(j-1)} = j^* + m e^{-\alpha j^*} \cdot \frac{e^{\alpha}}{1 - e^{-\alpha}}.
\end{aligned}$$

Since $j^* \geq \frac{2 \ln m}{\alpha} = \frac{8 \ln^2 m}{1 - 1/e}$, we have $m e^{-\alpha j^*} \leq m e^{-2 \ln m} = \frac{1}{m}$, and $\frac{e^{\alpha}}{1 - e^{-\alpha}} \leq \frac{2}{\alpha} = \frac{8 \ln m}{1 - 1/e}$ for $m \geq 2$. Therefore, $\mathbb{E}[I] \leq j^* + \frac{1}{m} \cdot \frac{8 \ln m}{1 - 1/e} \leq \frac{9 \ln^2 m}{1 - 1/e}$ for high enough m. □

Lemma 4.6. $\Pr\left[I \geq \frac{9 \ln^2 m}{1 - 1/e}\right] \leq \frac{1}{m}$.

Proof. Let $\alpha = \frac{1-1/e}{4\ln m}$ and $j^* = \frac{9\ln^2 m}{1-1/e}$. As shown in the proof of Lemma 4.5, $\Pr\{|E(H_j)| \geq 1\} \leq me^{-\alpha j}$. Setting $j = j^* \geq \frac{2\ln m}{\alpha} = \frac{8\ln^2 m}{1-1/e}$ gives $me^{-\alpha j^*} \leq me^{-2\ln m} = \frac{1}{m}$. Hence, $\Pr[I \geq j^*] = \Pr\{|E(H_{j^*})| \geq 1\} \leq \frac{1}{m}$. □

We get that the procedure executes at most $O(\log^2 m)$ iterations with high probability, so the running time of the algorithm is polynomial with high probability. It remains to show, that with high probability, the cost of the solution returned by the algorithm is at most $O(\log^2 m) \cdot \texttt{OPT}$. We have:

Lemma 4.7. *With high probability, we have that* $\mathrm{cost}(\lambda_A) \leq \frac{27\ln^2 m}{1-1/e} \cdot \mathit{OPT}$.

Proof. Fix an expensive vertex $v \in V$; cheap vertices have cost at most $1 \leq 2\texttt{OPT}$ regardless of I. Let $\mu_v = \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$.

By $X^j_{v,i}$ denote the indicator that $x_{i,v} \geq t_i$ in iteration j. Since different iterations draw independent fresh thresholds, the cost variables $\{\mathrm{cost}\left(v, \lambda^j_A\right)\}_{j\geq 1}$ are i.i.d. with mean μ_v, and I is independent of the sequence $\left(\mathrm{cost}\left(v, \lambda^j_A\right)\right)_{j\geq 1}$. Applying Wald's identity:

$$\mathbb{E}\left[\mathrm{cost}(v, \lambda_A)\right] = \mathbb{E}[I] \cdot \mu_v \leq \frac{9\ln^2 m}{1-1/e} \cdot \mu_v.$$

By the weighted Chernoff bound with $\delta = 2$:

$$\Pr\left[\mathrm{cost}(v, \lambda_A) \geq \frac{27\ln^2 m}{1-1/e} \cdot \mu_v\right] \leq \left(\frac{e^2}{27}\right)^{\frac{9\ln^2 m}{1-1/e}\cdot\mu_v} \leq e^{-\frac{9\ln^2 m}{1-1/e}\cdot\mu_v} \leq e^{-\frac{9\ln^2 m}{1-1/e}} \leq \frac{1}{m^9},$$

where the last inequality uses $\frac{9\ln^2 m}{1-1/e} \geq 9\ln m$ for $m \geq 2$. By the union bound over all $n \leq m$ expensive vertices, with probability at least $1 - \frac{1}{m^8}$, for every $v \in V$:

$$\mathrm{cost}(v, \lambda_A) \leq \frac{27\ln^2 m}{1-1/e} \cdot \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \leq \frac{27\ln^2 m}{1-1/e} \cdot \texttt{OPT}.$$

□

We conclude, that with high probability the algorithm returns a solution which covers all edges in H in at most $O(\log^2 m)$ iterations and has cost at most $O(\log^2 m) \cdot \texttt{OPT}$. Since H connects every group U_r internally, the solution returned by the algorithm is a valid solution to the Connectivity problem with cost at most $O(\log^2 m) \cdot \texttt{OPT}$ with high probability. □

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A Weighted Chernoff Bound Derivation

Theorem A.1 ((weighted) Chernoff bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables taking values in $[0,1]$, let $w_1, w_2, \ldots, w_n$ be non-negative weights, such that for every $i \in [n]$ we have that $w_i \in [0,1]$. Let $X = \sum_{i=1}^{n} w_i \cdot X_i$ and let $\mu = \mathbb{E}[X]$. Then for every $\delta_1 > 0$ and $\delta_2 \in (0,1)$ we have that:*

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu} \qquad \Pr(X \leq (1-\delta_2)\cdot\mu) \leq \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

Proof. We first prove the upper-tail bound for $\delta_1 > 0$. For any $\lambda > 0$, Markov's inequality gives:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) = \Pr\Big(e^{\lambda X} \geq e^{\lambda\cdot(1+\delta_1)\cdot\mu}\Big) \leq \frac{\mathbb{E}\big[e^{\lambda X}\big]}{e^{\lambda\cdot(1+\delta_1)\cdot\mu}}.$$

By independence, we have $\mathbb{E}\big[e^{\lambda\cdot X}\big] = \prod_{i=1}^{n}\mathbb{E}\big[e^{\lambda\cdot w_i\cdot X_i}\big]$. Fix $i \in [n]$. Since $w_i \cdot X_i \in [0,1]$, by convexity of $y \mapsto e^{\lambda\cdot y}$ on $[0,1]$, for every $y \in [0,1]$, $e^{\lambda\cdot y} \leq (1-y)\cdot e^0 + y\cdot e^{\lambda} = 1 + \big(e^{\lambda}-1\big)\cdot y$. Substituting $y = w_i \cdot X_i$, taking expectation, and using $1 + t \leq e^t$, we obtain:

$$\mathbb{E}\Big[e^{\lambda\cdot w_i\cdot X_i}\Big] \leq 1 + \Big(e^{\lambda}-1\Big)\cdot w_i\cdot\mathbb{E}[X_i] \leq \exp\Big(\Big(e^{\lambda}-1\Big)\cdot w_i\cdot\mathbb{E}[X_i]\Big).$$

Hence:

$$\mathbb{E}\Big[e^{\lambda X}\Big] \leq \exp\left(\Big(e^{\lambda}-1\Big)\cdot\sum_{i=1}^{n} w_i\cdot\mathbb{E}[X_i]\right) = \exp\Big(\Big(e^{\lambda}-1\Big)\cdot\mu\Big).$$

Therefore, $\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp\big(\mu\cdot\big(e^{\lambda}-1-\lambda\cdot(1+\delta_1)\big)\big)$. The function $f(\lambda) = e^{\lambda}-1-\lambda\cdot(1+\delta_1)$ is minimized at $\lambda = \ln(1+\delta_1)$, and so:

$$\Pr(X \geq (1+\delta_1)\cdot\mu) \leq \exp(-\mu\cdot((1+\delta_1)\cdot\ln(1+\delta_1)-\delta_1)) = \left(\frac{e^{\delta_1}}{(1+\delta_1)^{1+\delta_1}}\right)^{\mu}.$$

Now let $\delta_2 \in (0,1)$. For any $t > 0$, again by Markov's inequality:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) = \Pr\Big(e^{-tX} \geq e^{-t\cdot(1-\delta_2)\cdot\mu}\Big) \leq \frac{\mathbb{E}\big[e^{-t\cdot X}\big]}{e^{-t\cdot(1-\delta_2)\cdot\mu}}.$$

Applying the same mgf estimate with $\lambda = -t < 0$ yields $\mathbb{E}\big[e^{-t\cdot X}\big] \leq \exp\big(\big(e^{-t}-1\big)\cdot\mu\big)$, so:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) \leq \exp\big(\mu\cdot\big(e^{-t}-1+t\cdot(1-\delta_2)\big)\big).$$

The right-hand side is minimized at $t = -\ln(1-\delta_2)$, which gives:

$$\Pr(X \leq (1-\delta_2)\cdot\mu) \leq \exp(-\mu\cdot(\delta_2+(1-\delta_2)\cdot\ln(1-\delta_2))) = \left(\frac{e^{-\delta_2}}{(1-\delta_2)^{1-\delta_2}}\right)^{\mu}.$$

□